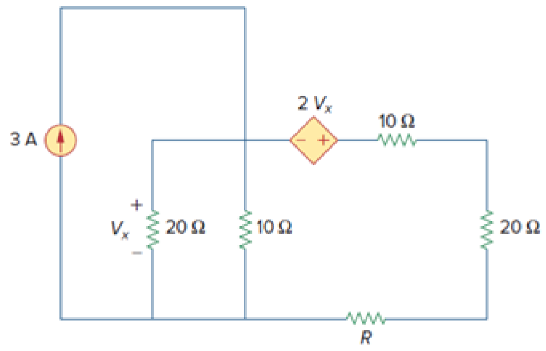
Chapter 4, Problem 70P

Problem

Determine the maximum power delivered to the variable resistor *R* shown in the circuit of Fig. 4.136.

Figure. 4.136:



Step-by-step solution

Step 1 of 11

Refer to Figure 4.136 in the text book.

Disconnect the load resistance to calculate the open circuit voltage.

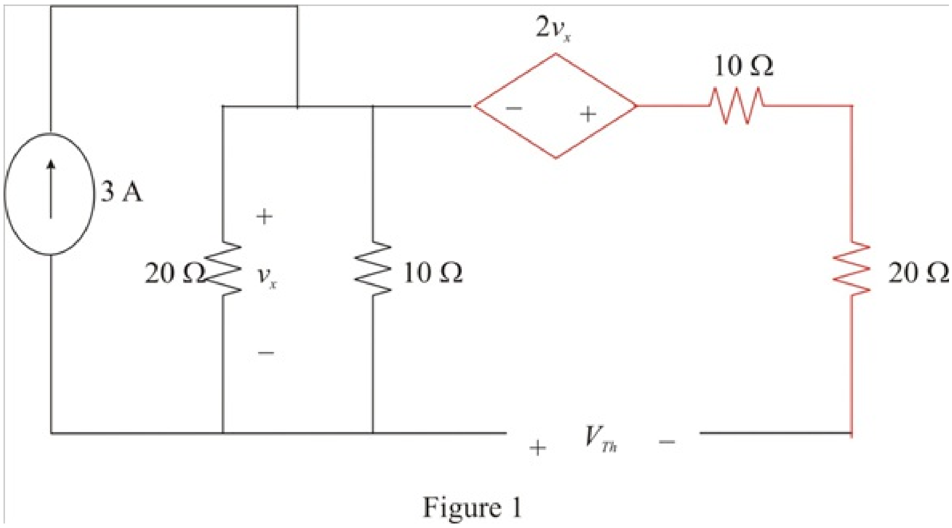


Figure 1

Step 2 of 11

Notice that no current passes through the components indicated in red color.

Calculate the value of the voltage v_x.

$$\begin{aligned} v_x &= (3)(20 \| 10) \\ &= (3)\left(\frac{(20)(10)}{20+10}\right) \\ &= \frac{600}{30} \\ &= 20 \text{ V} \end{aligned}$$

Therefore, he value of the voltage v_x is 20 V.

Step 3 of 11

Calculate the value of the Thevenin's voltage V_{Th}.

$$\begin{aligned} V_{\text{Th}} &= -2v_x - v_x \\ &= -3v_x \end{aligned}$$

Substitute $20\ \text{V}$ for v_x.

$$\begin{aligned} V_{\text{Th}} &= -3(20) \\ &= -60\ \text{V} \end{aligned}$$

Therefore, the value of the Theenin's voltage V_{Th} is $-60\ \text{V}$.

Step 4 of 11

Short the load terminals to calculate the short circuit current.

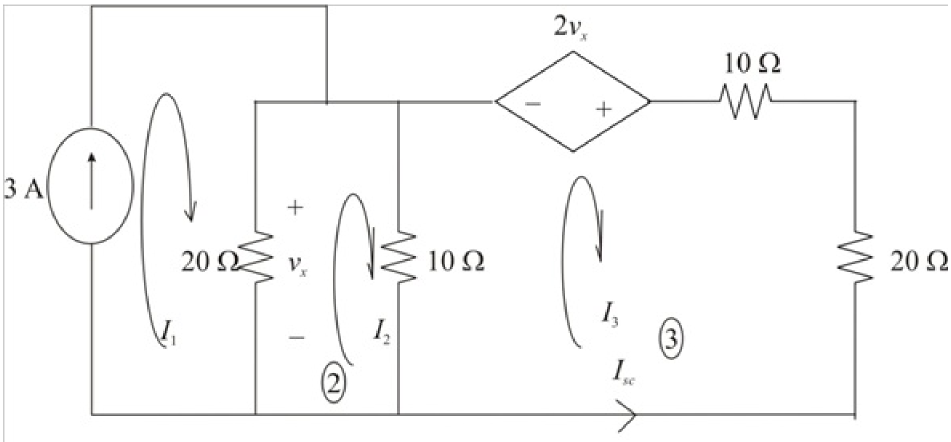


Figure 2

Step 5 of 11

Notice that the value of the current I_1 is $3\ \text{A}$.

Consider the expression for the voltage v_x.

$$v_x = (I_1 - I_2)(20)$$

Therefore, the value of the voltage v_x is $(I_1 - I_2)(20)$.

Step 6 of 11

Apply Kirchhoff's voltage law in loop-2.

$$\begin{aligned} &(I_2 - I_1)(20) + (I_2 - I_3)(10) = 0 \\ &20I_2 - 20I_1 + 10I_2 - 10I_3 = 0 \\ &-20I_1 + 30I_2 - 10I_3 = 0 \\ &30I_2 = 20I_1 + 10I_3 \end{aligned}$$

$$\begin{aligned} I_2 &= \frac{20I_1 + 10I_3}{30} \\ &= \frac{2I_1 + I_3}{3} \end{aligned}$$

Therefore, the expression for the current I_2 is $\frac{2I_1 + I_3}{3}$.

Apply Kirchhoff's voltage law in loop-3.

$$(I_3 - I_2)(10) - 2v_x + I_3(10+20) = 0$$
$$-10I_2 - 2v_x + I_3(10+20+10) = 0$$
$$-10I_2 - 2v_x + 40I_3 = 0$$

Substitute $(I_1 - I_2)(20)$ for v_x.

$$-10I_2 - 2(I_1 - I_2)(20) + 40I_3 = 0$$
$$-10I_2 - 40I_1 + 40I_2 + 40I_3 = 0$$
$$30I_2 - 40I_1 + 40I_3 = 0$$
$$3I_2 - 4I_1 + 4I_3 = 0$$

Substitute $\frac{2I_1 + I_3}{3}$ for I_2.

$$3\left(\frac{2I_1 + I_3}{3}\right) - 4I_1 + 4I_3 = 0$$
$$2I_1 + I_3 - 4I_1 + 4I_3 = 0$$
$$5I_3 - 2I_1 = 0$$
$$5I_3 = 2I_1$$
$$I_3 = \frac{2I_1}{5}$$

Substitute 3 A for I_1.

$$\begin{aligned} I_3 &= \frac{2(3)}{5} \\ &= \frac{6}{5} \\ &= 1.2\text{ A} \end{aligned}$$

Therefore, the value of the current I_3 is, 1.2 A.

Calculate the value of the short circuit current I_{sc}.

$$I_{sc} = -I_3$$

Substitute 1.2 A for I_3.

$$I_{sc} = -1.2\text{ A}$$

Therefore, the value of the short circuit current I_{sc} is -1.2 A.

Calculate the value of the Thevenin's resistance R_{Th}.

$$R_{Th} = \frac{V_{Th}}{I_{sc}}$$

Substitute $-1.2\ \text{A}$ for I_{sc} and $-60\ \text{V}$ for V_{Th}.

$$\begin{aligned} R_{Th} &= \frac{-60}{-1.2} \\ &= 50\ \Omega \end{aligned}$$

Therefore, the value of the Thevenin's resistance R_{Th} is $50\ \Omega$.

For the maximum power to be delivered to the load, the value of the load must be equal to the Thevenin's resistance R_{Th}. Therefore, the value of the load R is $50\ \Omega$.

Step 11 of 11

Calculate the maximum power absorbed by the load.

$$p_{max} = \frac{{V_{Th}}^2}{4R}$$

Substitute $-60\ \text{V}$ for V_{Th} and $50\ \Omega$ for R.

$$\begin{aligned} p_{max} &= \frac{(-60)^2}{4(50)} \\ &= \frac{3600}{200} \\ &= 18\ \text{W} \end{aligned}$$

Therefore, the value of the maximum power absorbed by the load is $\boxed{18\ \text{W}}$.